

QINGDAO KRETTO TECHNOLOGY CO.,LTD.

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About us

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Company name

Qingdao Kretto Technology Co.,Ltd.



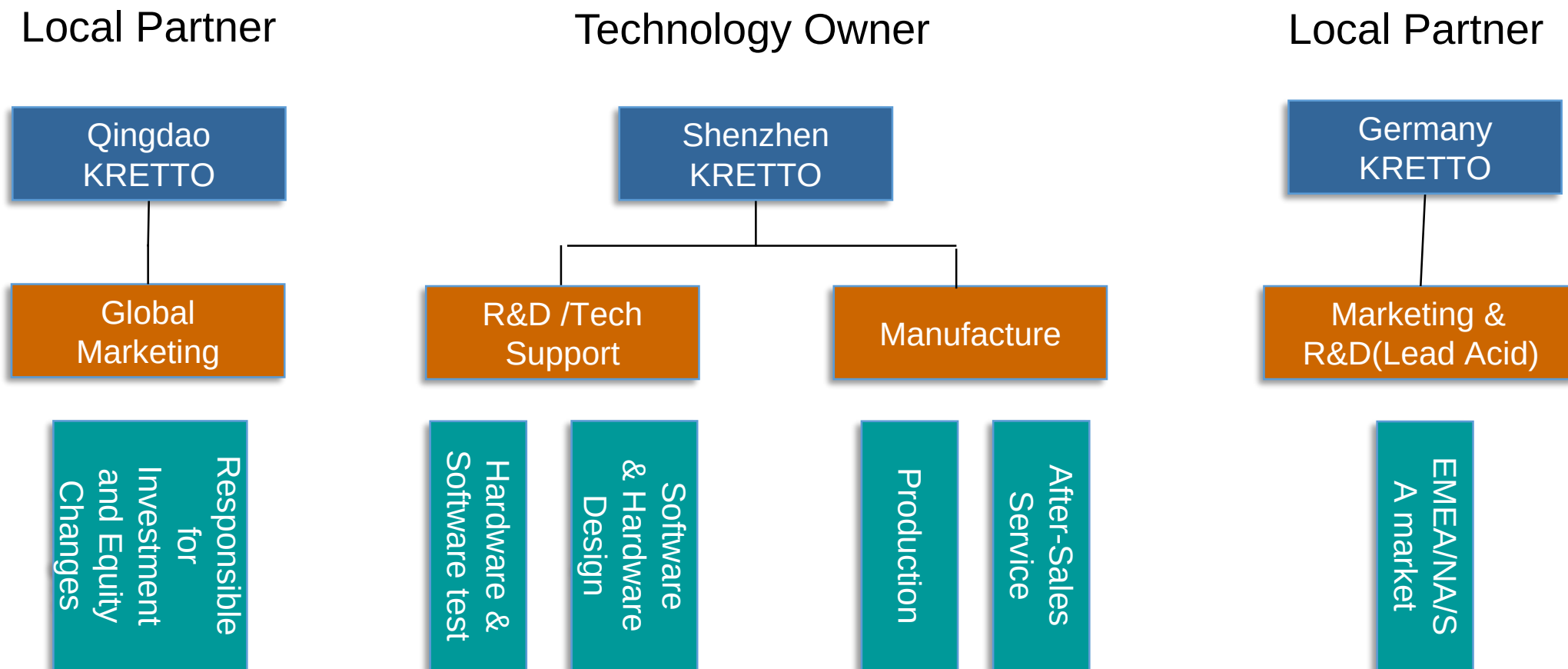
Founded

July 2012





KRETTO has senior technical personnel with long-term working experience and theoretical knowledge in the fields of batteries, electrical equipment, IT and artificial intelligence, and has formed a network with overseas partners in various countries with global business capabilities.



Dr. Jiang Jiuchun PhD

A leading figure in the power battery management industry, formerly a professor and doctoral advisor at Beijing Jiaotong University, currently serving as Chief Scientist at the Shenzhen Automotive Research Institute, Beijing Institute of Technology.

Recipient of the National High-Level Talent Special Support Program (Ten Thousand Talents Plan), China Automotive Industry Science and Technology Award (First Prize), China Machinery Industry Science and Technology Award (Special Prize), National Science and Technology Progress Award (Second Prize).

Dr. Yan Huiwen PhD

Ph.D. in Power Electronics from Tsinghua University.

Shenzhen Leading Talent.

Expert Reviewer for New Energy Projects.

EMBA from China Europe International Business School (CEIBS).

Former R&D Project Manager at Emerson

Chief Engineer at Kstar

Holds over 20 patents.

Currently an expert committee member and reviewer for the China Power Supply Society.

Dr. Wu Zhiqiang PhD

Senior Engineer, expert registered with the Ministry of Science and Technology, and special expert with the China Association for the Promotion of International Trade.

Long-term engaged in electric vehicle technology research, pioneered the “Three-Network Integration” technology framework and its application. Focuses on battery analysis technology and applications involving mechanism and data fusion.

Former expert member of the Electric Vehicle Standardization Committee in Guangdong Province and Shanghai.

Has led and participated in over 10 national key science and technology projects, and holds more than 10 authorized invention patents.

Dr. Jiang Yan PhD

Senior Engineer with a Doctoral Degree; Visiting Scholar at the University of California, San Diego from 2018 to 2019.

Involved in 10 projects, including National Natural Science Foundation and National Key R&D Program projects.

Published 14 SCI papers and holds 4 authorized invention patents.

Mainly responsible for the development of battery performance evaluation strategies.

Dr. Gao Yang PhD

Senior Engineer, responsible for core algorithm development, with a Doctoral Degree.

Involved in National Natural Science Foundation and National Key R&D Program projects.

Published over 10 SCI and EI indexed papers and holds 4 authorized invention patents.

Primarily responsible for the development of battery safety warning strategies.

Dr. Wei Shaoyuan PhD

Senior Engineer with a Doctoral Degree; Visiting Scholar at Chalmers University of Technology, Sweden, from 2019 to 2020.

Involved in nearly 10 National Key R&D Program projects and provincial/municipal-level projects. Published over 10 SCI and EI journal papers and holds 2 authorized invention patents.

Primarily responsible for capacity optimization and energy management strategies.

Lithium Lifeguard Battery Safety Solution BSS

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lithium battery intelligent solutions service provider

Early Warning System

Detects behavior of battery pack that leads to failure 12 to 24 hours ahead of failure

Utilizes a proprietary database of over 6000 BMS systems

Measures 20 different factors affecting battery safety

Universal Applicability

AI driven software can analyze unknown BMS and adapt software automatically

Analyzes all chemistries, LCO, NCM, LFP, etc.

User Information by App or Custom solution using commercial 4G or 5G

Central Server



Local Edge Computer/Data-Collector

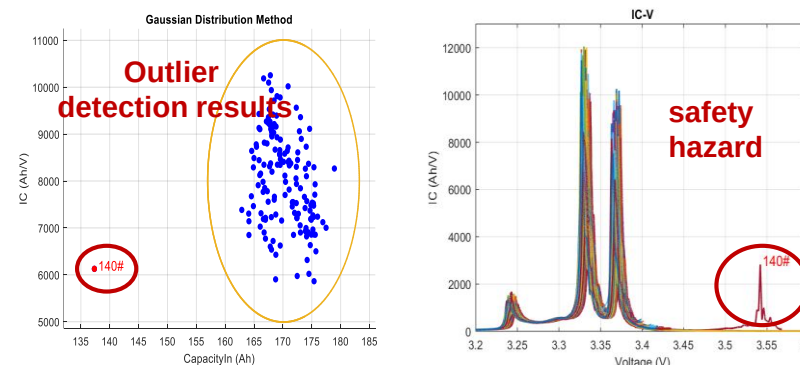


ESS Module

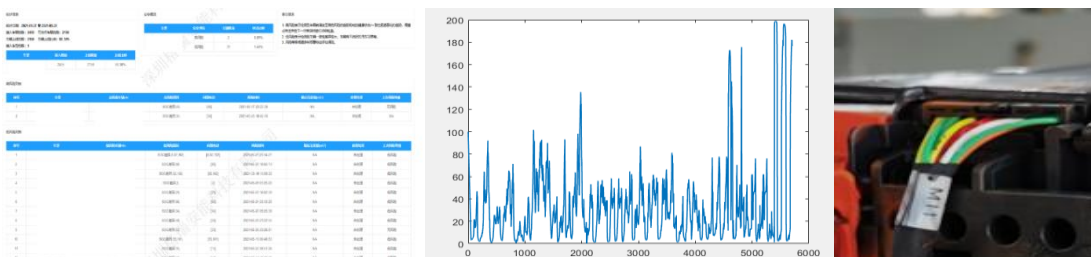
EV Module

For a long time, Qingdao KRETTO has systematically conducted battery test tests under the leadership of experts, and has accumulated huge battery behavior data through tests. Such a battery database is the foundation for KRETTO's AI-based BSS products and services to be overwhelmingly competitive in the market and achieve product differentiation.

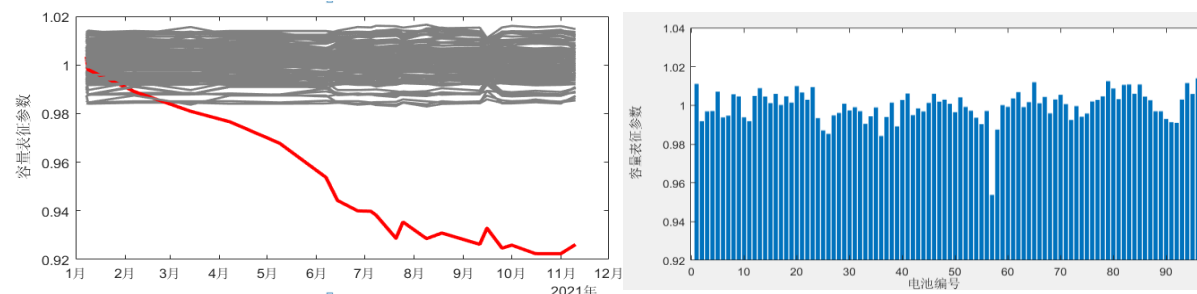
- ❖ Testing time: more than 10 years
- ❖ Testing place: China National Laboratory (participated by Chinese national scientists)
- ❖ Testing objects
 - Application product group: EV (passenger car, commercial vehicle, low speed electric vehicle, etc.), ESS
 - Battery type: prismatic, cylindrical, pouch cell
- ❖ Test quantity: More than 700,000 test vehicles (more than 700,000 battery packs)
- ❖ Test results: **more than 6,000 kinds of batteries operation data**



- ✓ Diagnose lithium plating in batteries,
- ✓ Warning of the possibility of thermal runaway.



- ✓ Periodically assess battery status, identify battery defects offline, and replace and maintain batteries in a timely manner.

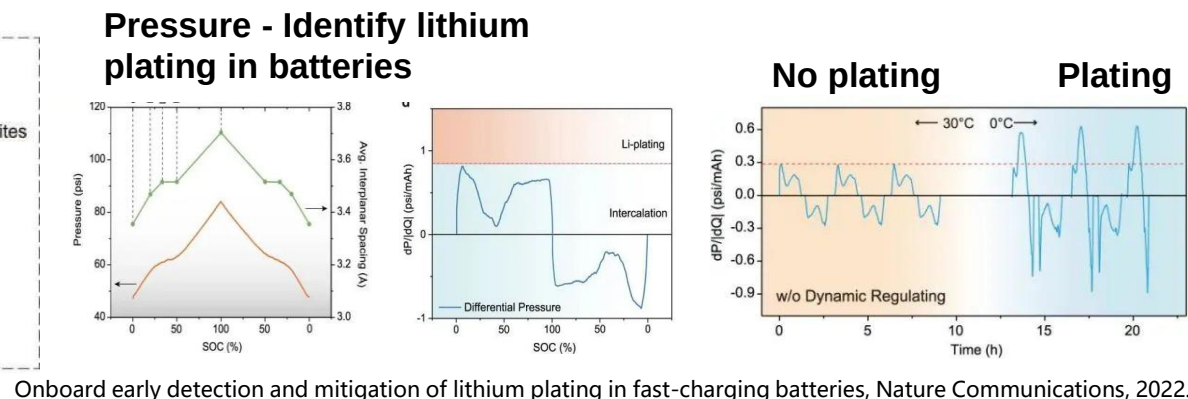
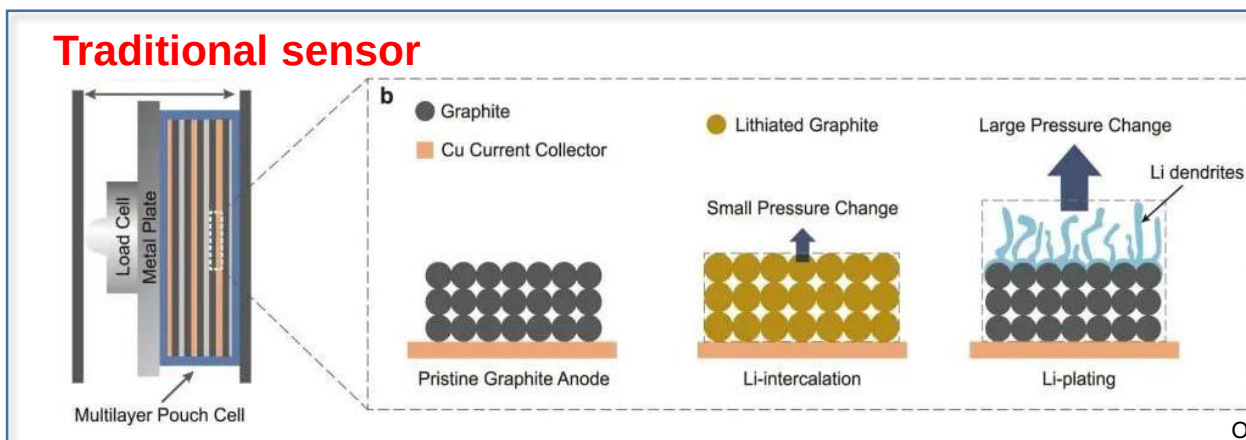
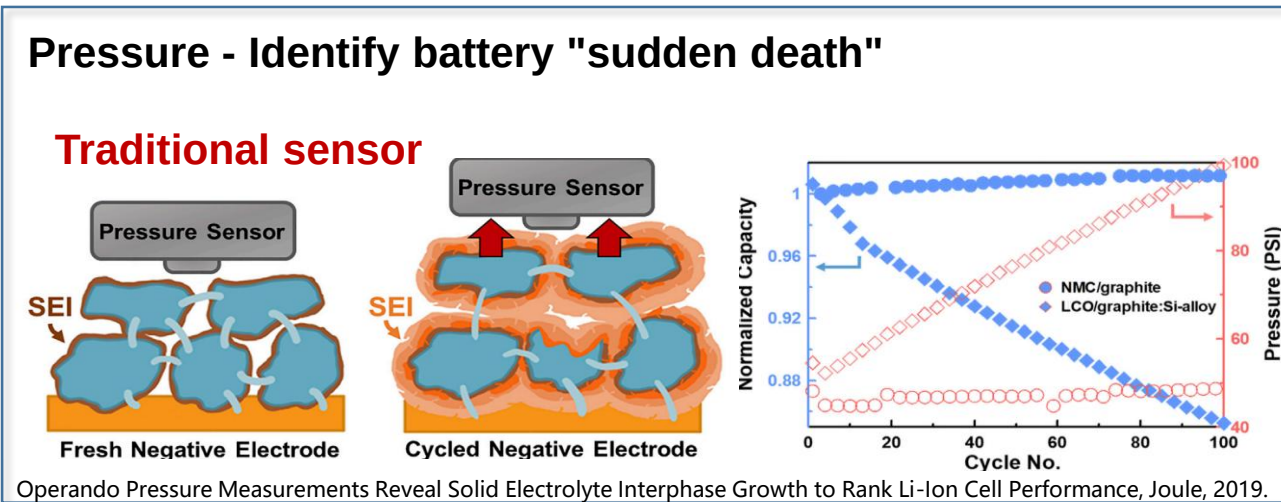
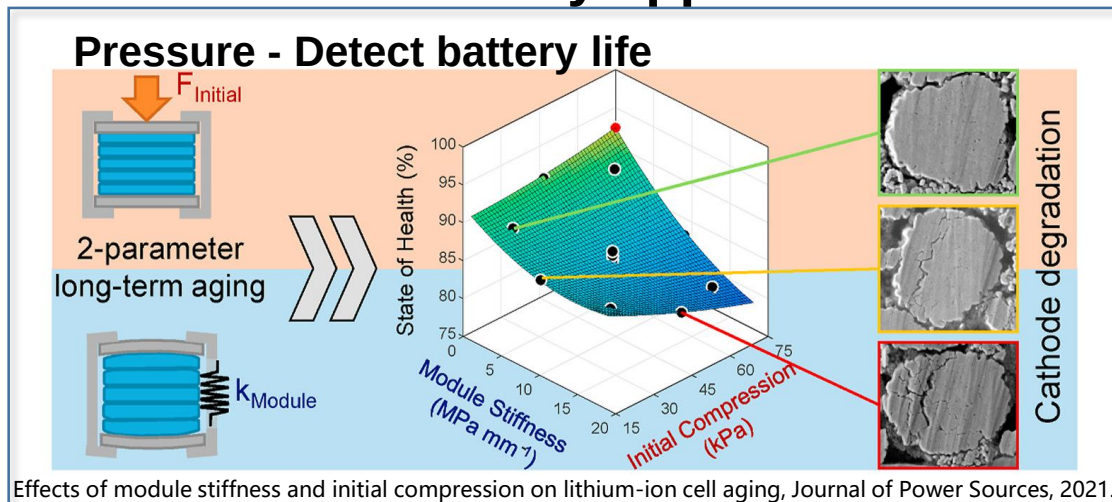


- ✓ Detecting the rapid decline in battery capacity, abnormal batteries can be replaced in advance to extend the life of the battery system.

Lithium Lifeguard Battery Safety Solution MINI BSS

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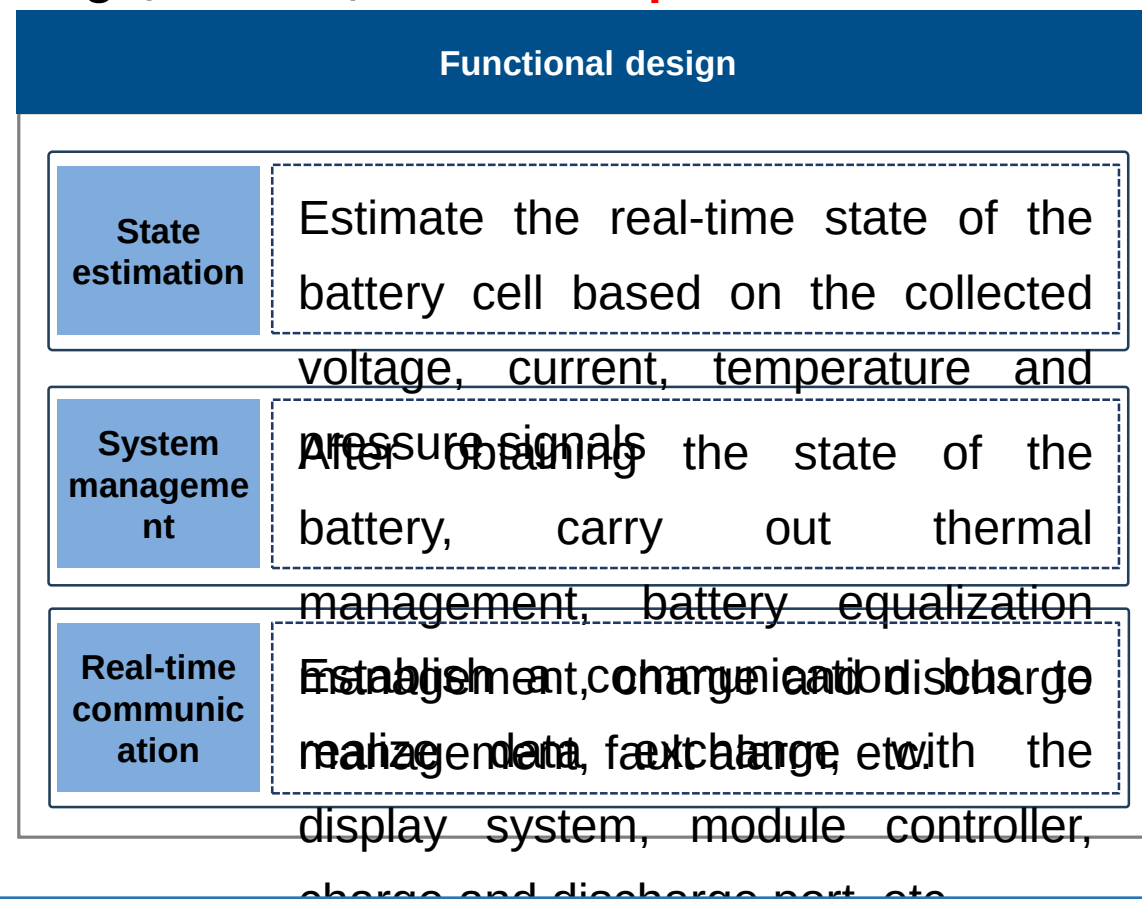
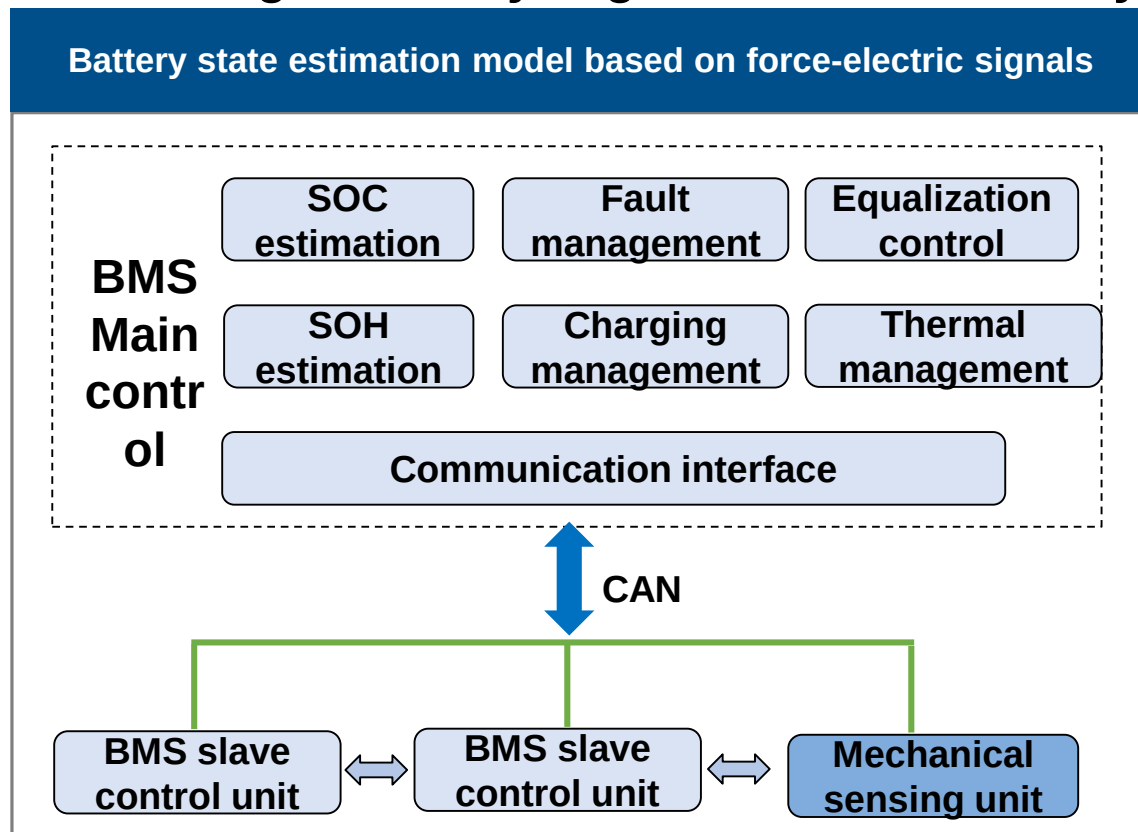
- ❑ The battery management method based on pressure is a **revolutionary change** in the field of battery applications.



Pressure detection has derived a variety of battery state estimation and fault diagnosis methods

□ The BMS architecture introducing mechanical sensing units

- Integrate **thin-film pressure sensors**, and estimate the battery SOC/SOH in real time by collecting and analyzing data such as battery voltage, current, and **in-situ pressure of the**



Introduce in-situ mechanical pressure signal to improve the estimation accuracy of SOX

1500V SESS-PCS Series Power Conversion System

4

Power Conversion System(PCS),In the energy storage system, PCS is connected between the battery system and the power grid (or load), and AC/DC conversion is carried out to achieve bidirectional conversion of electric energy.

- Overall efficiency up to 99%
- Small size, high power density
- Strong environmental adaptability



The maximum voltage on the DC side of the S
ESS-PCS series PCS is 1500V, and the models
include SESS-3450K-PCS, SESS-3150K-PCS, S
ESS-1725K-PCS, SESS-1575K-PCS, The power
rate is divided into 3450kW, 3150kW, 1725kW, 1
575kW, if you need other power can be replaced
by the above power derating.

SESS-3450K-PCS



PCS: Energy storage converter products

3450K: Power 3450KW

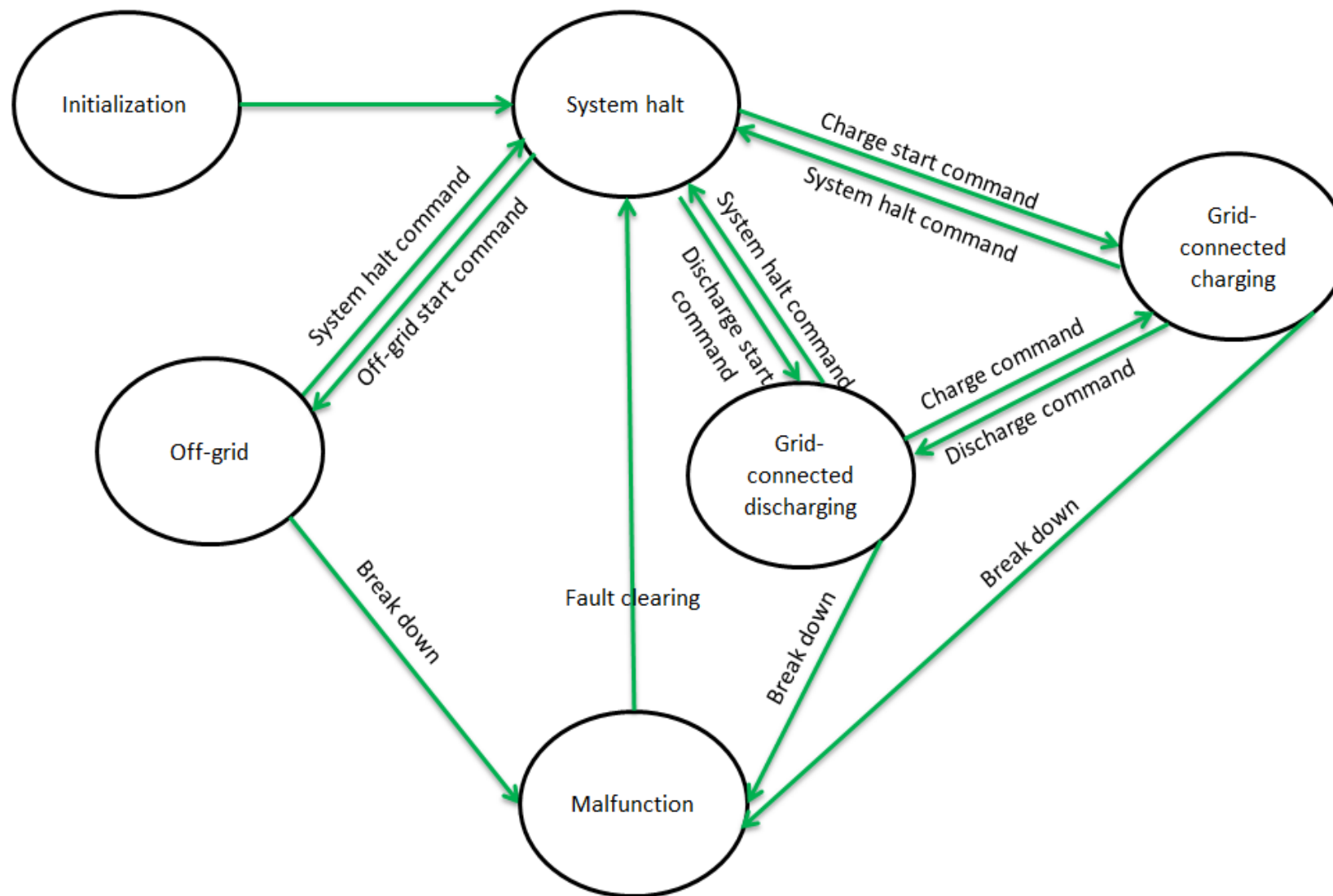
SESS: Energy storage series product code

Figure 1 Product naming rules

1) PCS technical specification

Table 1 Technical parameters of SESS-PCS series

Model		SESS-1575K-PCS	SESS-1525K-PCS	SESS-3150K-PCS	SESS-3450K-PCS
DC side parameter	DC voltage range	1000 ~ 1500V	1000 ~ 1500V	920 ~ 1500V	1000 ~ 1500V
	Rated DC power	1575kW	1725kW	3150kW	3450kW
	Max DC power	1732.5kW	1897.5kW	3465kW	3795kW
	DC rated current	1725A		3450A	
	Max DC current	1897A		3794A	
AC side parameter	Rated power	1575kW	1725kW	3150kW	3450kW
	Max output power	1732.5kW	1897.5kW	3465kW	3795kW
	Reactive power range	0 ~ 1575	0 ~ 1725	0 ~ 3150	0 ~ 3450
	Isolation mode	No isolation			
Grid-connected operation parameters	Rated grid voltage	690V	690V	630V	690V
	Allowable grid voltage	621V ~ 759V	621V ~ 759V	567V ~ 693V	621V ~ 759V
	Rated grid frequency	50Hz/60Hz			
	Allowable grid frequency	47Hz ~ 53Hz / 57Hz ~ 63Hz			
	Total harmonic distortion Rate of current	3% (rated power)			
	Power factor	-1 ~ 1			
	Charge and discharge conversion time	80ms			
Off-grid operation parameters	Rated output voltage	690V	690V	630V	690V
	Voltage unbalance degree	2%, within short time of less than 4%			
	Total harmonic distortion rate of voltage	3%(no load or rated resistive load)			
	Dynamic voltage transient range	10%(When the load increases from 20% to 100% or decreases from 100% to 20% under resistive load/balanced load conditions)			
	Output overvoltage protection value	Settable			
	Output undervoltage protection value	Settable			



Lithium Battery Testing System

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(1) 0V-20V,60A

(2) 0V-100V,60A

(3) 10V-200V,200A

Where to use

Automotive lead-acid battery, forklift battery and communication battery, industrial battery, low speed EV and lithium battery module

Battery type

Lead acid battery
(PE and AGM car battery)

Main Features

- Charging & Discharging testing
- DC internal resistance measurement
- Auxiliary data acquisition channel for voltage and temperature
- Channel parallel output power
- The third party equipment integration



01 0V-20V, 60A

02 0V-100V, 60A

03 10V-200V, 200A

DC 750V/±600A/160KW 2CH

1200V/1000A/C2-T (600KW)

1. Large Capacity battery Charge & Discharge System (DC 750V/±600A/160KW 2CH)

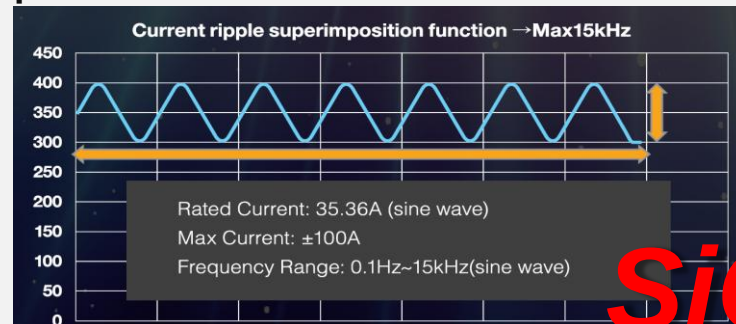
Battery Pack tester- Lithium Pack Testing Equipment

High Speed Current Response

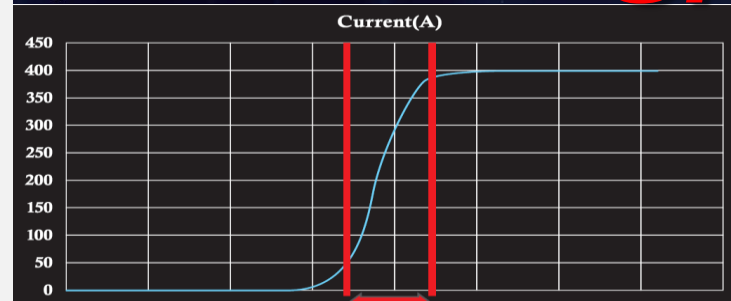
Current Ripple Superposition Function.

All SiC Small Size High Efficiency.

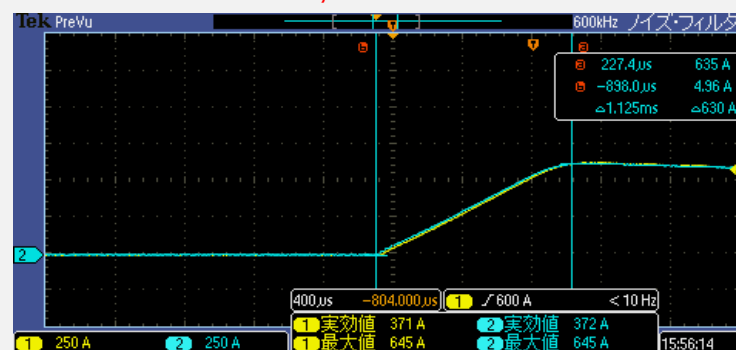
Max 4CH parallel connection possible



SiC



0A→600A, in 1msec 10%→90%



PN short out 0A→1200A, in 1msec at 10%→90%

01 0V-20V, 60A

02 0V-100V, 60A

03 10V-200V, 200A

DC 750V/±600A/160KW 2CH

1200V/1000A/C2-T (600KW)

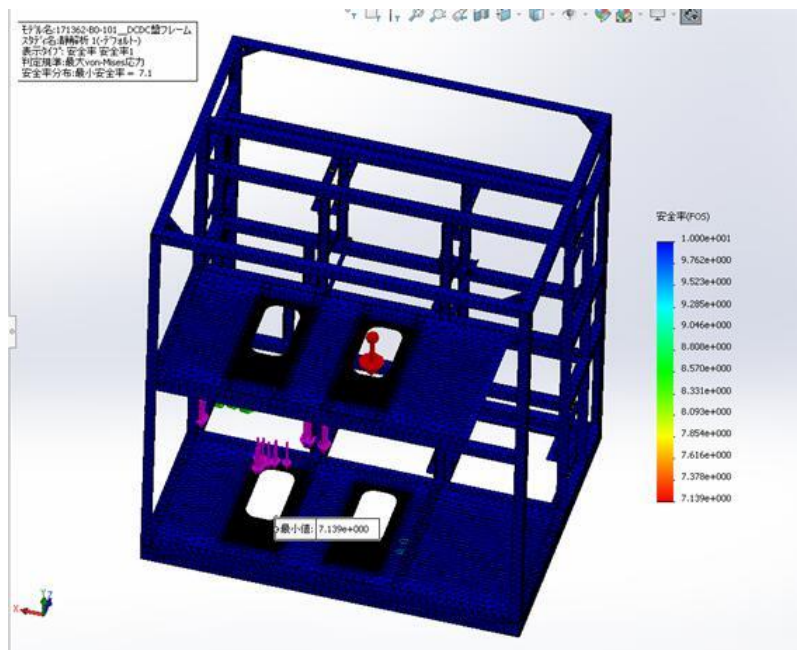
1. Large Capacity battery Charge & Discharge System(DC 750V/±600A/160KW 2CH)

Current	Item		Specification	Remarks
Power Supply AC Side	Power Factor		0.98 or more	AC Voltage THD ≤ 1% at rated power
	THD of Current		5% or less	
	Each Order Harmonic Distortion of Current		Each 3% or less	
DC Side (1CH)	Voltage Accuracy		±0.05% or F.S	Resistance Load
	Voltage Ripple		0.1%rms of 750V	
	Current Accuracy		±0.05% of F.S	
	Current Ripple		0.4%pp of 600A.(0.2%rms of 600A)	
	Power Accuracy		±0.1% of F.S	
	CV Mode	Voltage Setting Range	30V~750V (Resolution 10mV)	
	CC Mode	DC Current Setting Range	-600A~600A (Resolution 10mA)	
		AC Current Setting Range	-50A~50A (Resolution 10mA)	
		DC+AC Setting Range	-600A~600A (Resolution 10mA)	
	CP Mode	Power Setting Range	-160kW~160kW (Resolution 10W)	
	Operation with Charge/discharge Software		10msec pattern operation	
	High-speed Current Response		3msec or less	-90%→90% at change
	Total Efficiency		88% or more	at Rated Output Power

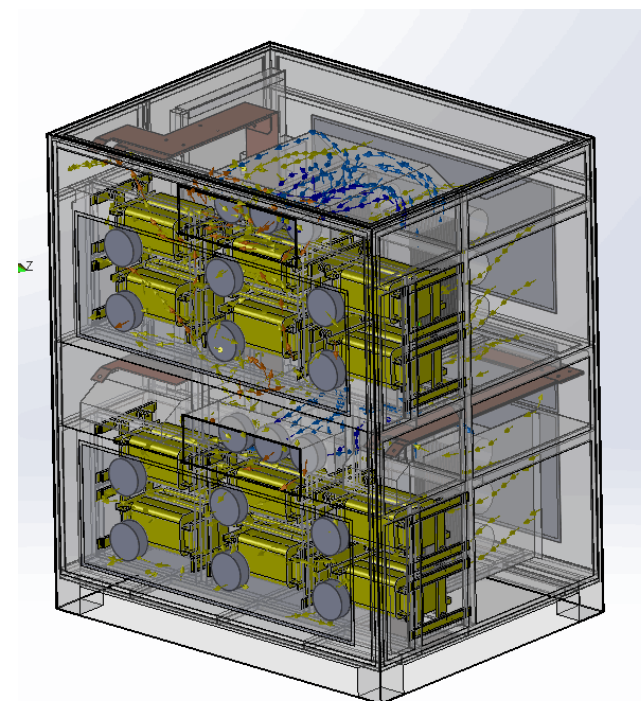
※ 1Accuracy guarantee : Outside Temperature Range : +23℃±5℃ Humidity : 70% or less

2. Large Capacity battery Charge & Discharge System (1200V/1000A/C2-T(600KW)

following testing equipment is used for Honda automobile battery pack testing. Heat dissipation simulation and structural stress simulation is always strictly carried out during each equipment design.



Structural stress simulation



Heat dissipation and airflow simulation

BMS design and production

6



Table 1. Specifications Table

No.	Technical requirements	Specifications
1	Single pack voltage	166.4V (52S)
2	Single pack capacity	26.62kWh
3	System capacity	53.24kWh (2 packs in series)
4	Specific energy	≥145Wh/Kg
5	Protection level	IP68
6	Dimensions	960mm*450mm*380mm
7	Thermal management	Liquid cooling/PTC heating
8	The number of series and parallel of the system	1P104S (2 packs in series)



Fig. 2 Battery pack outline drawing

Table 2. Technical indicators of liquid cooling system

No.	Item	Target value
1	Heat dissipating performance	Temperature rise < 8 °C @ 1 CHG, 25 °C
2	Heating rate	Heating rate ≥0.7°C/min@ Inlet temperature ≤50°C
3	Heating efficiency	≥60%@ Inlet temperature 15~35°C
4	Refrigeration efficiency	≥70% @ 1C charge

Note: Heating efficiency refers to the ratio of the heat required by the cell to the heat transferred by the total electric energy;
 Refrigeration efficiency refers to the ratio of the heat value taken away by the liquid to the total heat produced by the cell.

Table 3. Specifications

No.	Technical requirements	Specifications
1	Voltage	384V（120S Single pack）
2	Capacity	40.3kWh
3	Specific energy	≥140Wh/Kg
4	Protection level	IP68
5	Dimensions	1135mm*850mm*225mm
6	Thermal management	Liquid cooling/PTC heating



Fig. 8 Battery pack outline drawing

Table 4. Technical indicators of liquid cooling system

No.	Item	Target value
1	Heat dissipating performance	Temperature rise < 8 °C @ 1 CHG, 25 °C
2	Heating rate	Heating rate ≥0.7°C/min@ Inlet temperature ≤50°C
3	Heating efficiency	≥60%@ Inlet temperature 15~35°C
4	Refrigeration efficiency	≥70% @ 1C charge

Note: Heating efficiency refers to the ratio of the heat required by the cell to the heat transferred by the total electric energy;
 Refrigeration efficiency refers to the ratio of the heat value taken away by the liquid to the total heat produced by the cell.

Table 9. Specifications

No.	Item	Parameter		
System Parameter				
1	Rated Grid Voltage	120V	230V	400V
2	Allowable Grid Voltage	120V±10%	230V±10%	400V±10%
3	Rated Frequency	60Hz	50Hz	50Hz
4	Allowable Grid Frequency	60Hz±5%	50Hz±5%	50Hz±5%
5	Harmonics	< 5%	< 5%	< 5%
6	System Rated Power	500kW	500kW	500kW
7	System Rated Capacity	2534kWh	2534kWh	2534kWh
8	Protection Type	IP54		
9	Noise	< 75dB		
Environmental Parameter				
10	Temperature Range	-20°C~50°C		
11	Humidity	0-95%		
12	Installation Altitude Above Sea Level	< 1000m		
13	10 Feet Container Size	12.190m×2.438m×2.591m		
14	10 Feet Container Weight	21.4t		



Fig. 25 Schematic diagram

Table 10. Protection board function

No.	Item	Parameter
Protection		
1	Short-circuit Protection	Yes
2	Overload Protection	Yes
3	Battery Overvoltage Protection	Yes
4	Battery Low Voltage protection	Yes
5	AC Over/Under Voltage Protection	Yes
6	AC Over/Under Frequency Protection	Yes
7	Over Temperature Protection	Yes
8	Communication Failure	Yes

Lead Acid Battery Technology

7



Key Technology Advantages and Feature

HPL is based on the concept of **uniform consistent performance for each plate, each cell, and each battery.** HPL technology assures high quality, high efficiency, low-cost manufacturing and fast delivery. Using HPL Equipment batteries can be produced in less then **TWO-DAYS.**

High Speed ConCast Equipment, suitable for all kinds of Alloy Lead Positive & Negative Grid

40m/min

Effective Orifice Pasting Technology, Tiny Weight Tolerance, over-pasting thickness uniform in double sides.

40m/min

Pure Lead or Pb-Sn Alloy Lead

Thin Grid with frame

**Integrated
Solution for
2-Days
Delivery**

4 Hours

High Speed Pressure Curing Technology

Quality Improved
Cost reduced

6 Hours

High Speed Vacuum Formation Technology

Guarantee of Success

**Individual Equipment for process improvement
or
plant-wide integrated solution.**

Concast

1. Full-Frame Positive and Negative grids
2. Calcium Positive + Pure lead,
3. Grid Thickness:0.5mm.
4. Speed:40m/minute
5. No scrap
6. Energy savings
7. Cheapest full frame positive



Unbelievable: 4 Hour High Speed Curing Technology

- Energy Saving/Environmentally friendly
- Suitable for all kinds of battery: Flooded and AGM
- Curing time reduced $>80\%$ (4H VS 24H)
- Reduce lead oxide costs by 7% with higher porosity
- Uniform curing of all plates in the chamber
- Reduce energy costs by 35%(3BS) to 50%(4BS)



High speed formation**6 HOURS !!!
HIGH-SPEED FORMATION**

Power saving 30%

Formation factor 1.5 vs. > 2 for water bath or acid circulation

Time saving 70%

Fast formation in 4 – 6 hours vs 24 - 72 hours water bath

Water saving > 90%

No water evaporation loss – water recycled

One shot formation

No need to chill acid or use vacuum filling

No acid fog - No need for ventilation

All vapors are trapped and condensed



Thank You

